

Honours Analysis 1 Tutorials (Sec. 003)

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Please note that this document has not been proofread and may contain typos. Feel free to see or email me if you have any questions.

ANNOUNCEMENT: Due to the scheduling of a seminar, I would like to move my Thursday office hours to the 4:00pm-5:00pm slot. Please let me know if this causes any scheduling issues.

1 Seventh Tutorial

We begin by reiterating the definition of a limit for sequences. Recall that a sequence is a function $x : \mathbb{N} \rightarrow \mathbb{R}$, where we denote by x_n the value $x(n)$. It has become convention to denote a sequence by either (x_n) or $\{x_n\}_{n \in \mathbb{N}}$, I prefer the former. Intuitively, we like to picture a sequence as an “infinite list” of numbers, that may have repetitions. For this reason, it is important not to confuse a sequence with a set of real numbers, as sets do not allow for repetitions.

Definition 1. Let (x_n) be a sequence and fix $x \in \mathbb{R}$. We say that x_n *converges* to x (as $n \rightarrow \infty$) if, for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that $|x_n - x| < \varepsilon$ for all $n \geq N$. In symbolic terms:

$$(\forall \varepsilon > 0) (\exists N \in \mathbb{N}) (n \geq N \implies |x_n - x| < \varepsilon).$$

If this is the case, then we write $x_n \rightarrow x$, $x_n \xrightarrow{n \rightarrow \infty} x$, or

$$\lim x_n = x.$$

To help solidify the definition of convergence, we give a few explicit examples of convergent sequences. Although in practice one does not argue directly from the definition, it is important to fully understand the logic behind this definition.

Example 1. Show that

$$\lim_{n \rightarrow \infty} \frac{2n}{n+1} = 2.$$

Proof. Let $\varepsilon > 0$ be given, we must find $N \in \mathbb{N}$ such that

$$\left| \frac{2n}{n+1} - 2 \right| < \varepsilon, \quad \forall n \geq N.$$

First, we calculate for all $n \in \mathbb{N}$:

$$\left| \frac{2n}{n+1} - 2 \right| = \left| \frac{2n - 2(n+1)}{n+1} \right| = \frac{2}{n+1} \leq \frac{2}{n}.$$

By the Archimedean property, we can find $N \in \mathbb{N}$ such that $N > \frac{2}{\varepsilon}$. Then, for all $n \geq N$ there holds

$$\frac{2}{n} \leq \frac{2}{N} < 2 \cdot \frac{\varepsilon}{2} = \varepsilon.$$

Hence, if $n \geq N$, we see that

$$\left| \frac{2n}{n+1} - 2 \right| \leq \frac{2}{n} < \varepsilon,$$

which is what had to be shown. □

Example 2. Prove that

$$\lim_{n \rightarrow \infty} \frac{3n+1}{2n+5} = \frac{3}{2}.$$

Proof. Let $\varepsilon > 0$ be given, we seek $N \in \mathbb{N}$ such that

$$\left| \frac{3n+1}{2n+5} - \frac{3}{2} \right| < \varepsilon$$

for all $n \geq N$. Naturally, for each $n \in \mathbb{N}$, we have the following estimate:

$$\begin{aligned} \left| \frac{3n+1}{2n+5} - \frac{3}{2} \right| &= \left| \frac{2(3n+1) - 3(2n+5)}{2(2n+5)} \right| = \left| \frac{2 - 15}{2(2n+5)} \right| \\ &= \frac{13}{2(2n+5)} \\ &\leq \frac{13}{2n} \\ &\leq \frac{13}{n}. \end{aligned}$$

Choose $N \in \mathbb{N}$ so large that $\frac{1}{N} < \frac{\varepsilon}{13}$ (why can we do this?), if $n \geq N$ the above implies that

$$\left| \frac{3n+1}{2n+5} - \frac{3}{2} \right| \leq \frac{13}{n} \leq \frac{13}{N} < 13 \cdot \frac{\varepsilon}{13} = \varepsilon.$$

By definition, this means that

$$\frac{3n+1}{2n+5} \xrightarrow{n \rightarrow \infty} \frac{3}{2}.$$

□

Remark 1.1. We would like to point out our general strategy for computing limits via the definition. If we wish to show that a sequence (x_n) converges to $x \in \mathbb{R}$, we start by fixing $\varepsilon > 0$. Our goal is then to show that there exists $N \in \mathbb{N}$ such that $|x_n - x| < \varepsilon$ whenever $n \geq N$. To *find* such an N , we estimate $|x_n - x|$ for **all** n . This is done in the hopes of bounding the (often complex) expression $|x_n - x|$ by some term, depending on n , that is easier to estimate.

Example 3. Prove that

$$\lim \frac{n^2 - 1}{2n^2 + 3} = \frac{1}{2}.$$

Proof. Following the outline in the remark, let $\varepsilon > 0$ and consider the expression:

$$\begin{aligned} \left| \frac{n^2 - 1}{2n^2 + 3} - \frac{1}{2} \right| &= \left| \frac{2(n^2 - 1) - (2n^2 + 3)}{2(2n^2 + 3)} \right| \\ &= \left| \frac{5}{2(2n^2 + 3)} \right| \\ &\leq \frac{4}{n^2}. \end{aligned}$$

Now, it will be easier to find our candidate for N . Let $N \in \mathbb{N}$ be such that

$$N > \frac{2}{\sqrt{\varepsilon}},$$

which we know to exist by the Archimedean property. If $n \geq N$, then our estimate above tells us that

$$\left| \frac{n^2 - 1}{2n^2 + 3} - \frac{1}{2} \right| \leq \frac{4}{n^2} \leq \frac{4}{N^2} < 4 \cdot \frac{\varepsilon}{4} = \varepsilon.$$

This is what had to be shown.

□

We give one final example.

Example 4. Prove that

$$\lim_{n \rightarrow \infty} \frac{\sqrt{n}}{n+1} = 0.$$

Proof. Let $\varepsilon > 0$ be given. For every $n \in \mathbb{N}$, we have

$$\left| \frac{\sqrt{n}}{n+1} - 0 \right| = \frac{\sqrt{n}}{n+1} \leq \frac{\sqrt{n}}{n} = \frac{1}{\sqrt{n}}.$$

Let now $N \in \mathbb{N}$ be such that $N > \frac{1}{\varepsilon^2}$. Then, $\frac{1}{\sqrt{N}} < \varepsilon$ so that, for all $n \geq N$,

$$\left| \frac{\sqrt{n}}{n+1} - 0 \right| \leq \frac{1}{\sqrt{n}} \leq \frac{1}{\sqrt{N}} < \varepsilon.$$

□

1.1 Other Properties and Limits

We note that the convergence of a sequence (x_n) does imply that convergence of the sequence $(|x_n|)$. Indeed, we prove this below.

Proposition 1. *Let (x_n) be a sequence of real numbers converging to $x \in \mathbb{R}$. Then, $|x_n|$ converges to $|x|$, as $n \rightarrow \infty$.*

Proof. Let $\varepsilon > 0$ be given, since $\lim x_n = x$, we can find $N \in \mathbb{N}$ so large that $|x_n - x| < \varepsilon$ for all $n \geq N$. By the reverse triangle inequality, for all $n \geq N$ there then holds

$$||x_n| - |x|| \leq |x_n - x| < \varepsilon.$$

Hence, $\lim |x_n| = |x|$.

□

The converse need not hold in general. Indeed, consider the sequence (x_n) given by $x_n := (-1)^n$. This sequence does **not** converge to any real number x , but the sequence given by $|x_n|$ is always equal to 1. Since constant sequences converge, this means that $(|x_n|)$ is convergent. Despite this, a partial converse to the above does hold.

Proposition 2. *Let (x_n) be a sequence of real numbers. If $(|x_n|)$ converges to 0 as $n \rightarrow \infty$, then so does (x_n) .*

Proof. Given $\varepsilon > 0$, we can find $N \in \mathbb{N}$ so large that $||x_n| - 0| < \varepsilon$, whenever $n \geq N$. But this means that $|x_n| < \varepsilon$ for all $n \geq N$. This proves the claim. $\square$

Theorem 1.1. *If $a > 1$ is a real number, then*

$$\lim \left(\frac{1}{a^n} \right) = 0.$$

Moreover, if $a \in (-1, 1)$, then $\lim a^n = 0$.

Proof. First suppose that $a > 1$. Then, $a = 1 + t$ for some $t > 0$. By Bernoulli's inequality, we know that

$$a^n = (1 + t)^n \geq 1 + nt, \quad \forall n \geq 1.$$

Therefore, for $n \in \mathbb{N}$, we have the estimate:

$$\left| \frac{1}{a^n} \right| = \frac{1}{a^n} \leq \frac{1}{1 + nt} \leq \frac{1}{nt}.$$

Choose $N \in \mathbb{N}$ such that $N > \frac{1}{t\varepsilon} > 0$. For all $n \geq N$, we have

$$\left| \frac{1}{a^n} - 0 \right| \leq \frac{1}{nt} \leq \frac{1}{Nt} < \frac{t\varepsilon}{t} = \varepsilon.$$

This shows that $\frac{1}{a^n} \rightarrow 0$, as $n \rightarrow \infty$. Now, we handle the case where $a \in (0, 1)$. Consider $b := \frac{1}{a} > 1$; for every $n \in \mathbb{N}$ we have

$$a^n = \frac{1}{b^n}.$$

By the first part, it follows that

$$\lim a^n = \lim \left(\frac{1}{b^n} \right) = 0.$$

If $a = 0$, then obviously $\lim a^n = 0$. Thus, it only remains to handle the case $a \in (-1, 0)$. In this case, note that $|a| \in (0, 1)$ and

$$|a^n| = |a|^n, \quad \forall n \in \mathbb{N}.$$

By the previous case (the case of $a \in (0, 1)$), we see that

$$\lim |a^n| = \lim (|a|^n) = 0.$$

Invoking Proposition 2, we see that $\lim a^n = 0$ in this case as well. With this, the proof is complete. $\square$

1.2 About the Ratio Test

We begin with the following theorem.

Theorem 1.2. *Let (x_n) be a sequence of **positive** real numbers such that*

$$L := \lim \left(\frac{x_{n+1}}{x_n} \right)$$

exists. If $L < 1$, then (x_n) converges to 0.

Proof. Fix $r \in (L, 1)$ and define $\varepsilon_0 := r - L$. Choose $N \in \mathbb{N}$ such that

$$\left| \frac{x_{n+1}}{x_n} - L \right| < \varepsilon_0, \quad \forall n \geq N.$$

Since $x_n > 0$ for all n , we know that $L \geq 0$. If $n \geq N$, then the above implies that

$$\frac{x_{n+1}}{x_n} = \left| \frac{x_{n+1}}{x_n} \right| \leq \left| \frac{x_{n+1}}{x_n} - L \right| + |L| < L + \varepsilon_0 = r.$$

Thus, for all $n \geq N$ there holds

$$0 < x_{n+1} < x_n r < x_{n-1} r^2 < \cdots < x_N r^{n-N+1} =: Cr^n,$$

for some constant $C > 0$ that does not depend on n . By Theorem 1.1, we know that $\lim r^n = 0$. In particular, we know that $\lim(Cr^n) = 0$. Thus, given $\varepsilon > 0$, we can find $N' \in \mathbb{N}$ so large that

$$Cr^n = |Cr^n| < \varepsilon, \quad \forall n \geq N'.$$

Let $K := \max(N, N')$. If $n \geq K + 1$, then

$$|x_n| \leq Cr^{n-1} < \varepsilon,$$

since $n - 1 \geq N'$. Thus, the theorem is proven. □

Let (x_n) be a sequence of real numbers. For every $N \in \mathbb{N}$, let us define

$$S_N := \sum_{n=1}^N x_n.$$

This gives us a *new* sequence (S_N) in $\mathbb{R}$. We then say the infinite series $\sum_{n=1}^{\infty} x_n$ converges if there exists a real number x such that

$$\lim_{N \rightarrow \infty} S_N = \lim_{N \rightarrow \infty} \sum_{n=0}^N x_n = x.$$

In this case, we will denote this limit by the quantity $\sum_{n=1}^{\infty} x_n$. The series $\sum_{n=1}^{\infty} x_n$ is said to be *absolutely convergent* if the series

$$\sum_{n=1}^{\infty} |x_n|$$

is convergent. We recall that an absolutely convergent series is convergent, however we shall not prove this here.

Proposition 3. *Let (x_n) be a sequence of non-zero real numbers such that*

$$\lim \left| \frac{x_{n+1}}{x_n} \right| = L$$

exists. If $L < 1$, then the series $\sum_{n=1}^{\infty} x_n$ is absolutely convergent.

Proof. It is automatic that $L \geq 0$. Let us fix $L < r < 1$ and define $\varepsilon_0 := r - L > 0$. There exists $N \in \mathbb{N}$ so large that

$$\left| \frac{x_{n+1}}{x_n} - L \right| < \varepsilon_0, \quad \forall n \geq N.$$

Therefore,

$$\left| \frac{x_{n+1}}{x_n} \right| \leq L + \left| \frac{x_{n+1}}{x_n} - L \right| < L + \varepsilon_0 = r$$

for all $n \geq N$. That is,

$$|x_{n+1}| \leq r |x_n| \quad \forall n \geq N.$$

It follows that $|x_{n+k}| \leq r^k |x_n|$ for all $k \geq 1$ and $n \geq N$. But then, for every $K \geq N + 1$, we see that

$$\begin{aligned} \sum_{n=N+1}^K |x_n| &\leq \sum_{k=1}^K |x_{N+k}| \leq \sum_{k=1}^K r^k |x_N| \leq |x_N| \sum_{k=1}^{\infty} r^k \\ &= |x_N| \frac{1}{1-r}. \end{aligned}$$

Therefore, the sequence

$$(S_K)_{K=1}^{\infty} = \left(\sum_{k=1}^K |x_k| \right)$$

is bounded. This implies (by the monotone convergence theorem - which you will all see soon) that the series $\sum_{n=1}^{\infty} |x_n|$ converges. $\square$

1.3 Some Additional Limit Proofs

We will use the remaining time to discuss additional “less obvious” limits. For these, we will simplify our argument by using the *squeeze* and limit theorems. First, we show that

$$\lim_{n \rightarrow \infty} \frac{2^n}{n!} = 0. \quad (1.1)$$

To do this, we will show that

$$0 < \frac{2^n}{n!} \leq 2 \left(\frac{2}{3} \right)^{n-2}, \quad \forall n \geq 2.$$

If this indeed true, then the squeeze theorem would imply that

$$\lim_{n \rightarrow \infty} \frac{2^n}{n!} = 0.$$

Thus, we are reduced to verifying the aforementioned inequality. For this, we shall argue by induction. As a base case, let us check the inequality for $n = 2$:

$$0 < \frac{2^2}{2!} = 2 \quad \text{and} \quad 2 \left(\frac{2}{3} \right)^0 = 2.$$

Our base case is thence satisfied. Now, for the inductive step. Assume that

$$0 < \frac{2^n}{n!} \leq 2 \left(\frac{2}{3} \right)^{n-2}$$

for some $n \geq 2$. Clearly, by the inductive hypothesis:

$$\begin{aligned} 0 < \frac{2^{n+1}}{(n+1)!} &= \frac{2}{(n+1)} \cdot \frac{2^n}{n!} \leq \frac{2 \cdot 2}{(n+1)} \left(\frac{2}{3} \right)^{n-2} \\ &\leq 2 \left(\frac{2}{3} \right)^{n-2+1} \\ &= 2 \left(\frac{2}{3} \right)^{(n+1)-2}. \end{aligned}$$

Here, we have used that $n \geq 2$ implies $n + 1 \geq 3$ and, consequently, $\frac{1}{n+1} \leq \frac{1}{3}$. By our earlier remarks (where we discussed the application of the squeeze theorem), we see that (1.1) holds.

Next, we discuss the following limit:

$$\lim \frac{n!}{n^n} = 0. \quad (1.2)$$

Again, this can be solved relatively easily by the use of the squeeze theorem. Given $n \in \mathbb{N}$, observe that

$$0 < \frac{n!}{n^n} = \frac{n \cdot (n-1) \cdots 2 \cdot 1}{\underbrace{n \cdot n \cdots n}_{n \text{ times}}} = \prod_{k=1}^n \frac{k}{n} \leq \frac{1}{n}.$$

This last inequality uses the fact that $\frac{k}{n} \leq 1$, for all $k = 2, \dots, n$. Thus, for every $n \in \mathbb{N}$, one has

$$0 \leq \frac{n!}{n^n} \leq \frac{1}{n}.$$

Since

$$\frac{1}{n} \xrightarrow{n \rightarrow \infty} 0,$$

the Squeeze Theorem shows that (1.2) is true.

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